

2014-2015 – Master 2 – Macro I

Lecture 10 : Competitive search equilibrium

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Version 1.0

29/09/2014

Disclaimer

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Introduction

- ▶ The idea behind competitive search equilibrium is that search intermediaries could arise and internalize the congestion externalities and appropriability problems associated with search.
- ▶ This is similar to the issues arising in local public good provision and coasian internalization of externalities through contracts and property rights.

1. Fishing Clubs

Model

- ▶ A pond where the number of fish being caught is $F(N)$, where N is the number of fishermen.
- ▶ Each fisherman catches $F(N)/N$ fish
- ▶ The value of each fish is 1 and the opportunity cost of fishing per fisherman is c .

1. Fishing Clubs

Equilibrium without clubs and Optimum

- ▶ The equilibrium number of fishermen is such that

$$\frac{F(N)}{N} = c.$$

- ▶ That is

Average productivity of fishing = cost of fishing

- ▶ But social welfare is

$$W(N) = F(N) - cN.$$

- ▶ Thus the optimum is

$$F'(N) = c.$$

- ▶ That is

Marginal productivity of fishing = cost of fishing.

- ▶ The average/marginal distinction is the essence of a congestion externality.

1. Fishing Clubs

Subponds

- ▶ The pond is cut into K identical subponds
- ▶ Assumption about fishing technology : if n people fish in a subpond, they catch $F(Kn)/K$ fish
- ▶ Consistency check : if N fishermen are uniformly distributed over the subponds, they catch $KF(Kn)/K = F(N)$ fish
- ▶ Each subpond is owned by a firm which charges an entry fee p per fisherman.
- ▶ These firms compete for customers but earn positive rents due to their ownership of part of the ponds
- ▶ Competition brings their prices in line, but a firm can charge a higher price and remain competitive if in exchange for that it offers less congestion, i.e. more fish, to its clients (members)

1. Fishing Clubs

Subponds

- ▶ The value to a customer of a pond which charges p and has n people is

$$v(p, n) = \frac{F(Kn)}{Kn} - p - c.$$

- ▶ In equilibrium, all subponds such that $n > 0$ (called "active") must yield the same value to the customers :

$$\frac{F(Kn)}{Kn} - p - c = \bar{u}.$$

- ▶ This generate an equilibrium relationship between the density of fishermen at an active pond and its entry fee :

$$p(n) = \frac{F(Kn)}{Kn} - c - \bar{u}. \quad (1)$$

1. Fishing Clubs

Profit maximization

- ▶ Since owners maximize profits, they will choose the density that maximizes $np(n)$:

$$n = \arg \max np(n) = n \left(\frac{F(Kn)}{Kn} - c - \bar{u} \right).$$

- ▶ The FOC of firms is $F'(Kn) = c + \bar{u}$.
- ▶ At a symmetrical equilibrium, $n = N/K$ and therefore : $F'(N) = c + \bar{u}$.
- ▶ Since there are many (many) potential fishermen that can freely join clubs, in equilibrium $\bar{u} = 0$.
- ▶ Therefore, $F'(N) = c$.
- ▶ The equilibrium with competitive clubs is efficient.

1. Fishing Clubs

Efficient equilibrium

- ▶ The price schedule defined by (1) accurately reflects the congestion externality : clubs with more fishermen must charge a lower price.
- ▶ This makes the total profit of a club equal to the economic surplus it creates, which leads to an efficient outcome.
- ▶ The theory of competitive search (Moene (1995)) applies these ideas to the Mortensen-Pissarides matching model.

2. Search Clubs

Assumptions

- ▶ There are several ways of doing it. Here we assume "wage posting".
- ▶ firms and workers can join a search intermediary, i.e. they *assign* their vacancy and candidacy to one of those clubs.
- ▶ Vacancies and unemployed workers are only matched together if they belong to the same club.
- ▶ Each club specifies the wage that must be paid to the worker after the match has taken place.
- ▶ A club contains vacancies and unemployed workers in proportions θ , and specifies the wage w that will be paid after a vacancy is matched with an unemployed.
- ▶ The unemployed members of the club are charged f per unit of time until they are matched, while vacancies are charged g per unit of time.

2. Search Clubs

Values

- ▶ **Note :** We restrict here to stationary equilibrium, so that all $\dot{V}$ are 0.
- ▶ In equilibrium, the value of being unemployed V_u which is the maximum value offered to the unemployed across clubs.
- ▶ V_v is the maximum of posting a vacancy across clubs. With free entry , $V_v = 0$.
- ▶ Let $V_u(w, \theta, f)$ the value to the unemployed of joining a club with (w, θ, f) . The asset valuation equation is

$$rV_u(w, \theta, f) = -f + \theta q(\theta)(V_e(w) - V_u(w, \theta, f)),$$

where $V_e(w)$ is the value of a job to the worker paying a wage w , which satisfies

$$rV_e(w) = w + s(V_u - V_e(w)).$$

2. Search Clubs

Equilibrium

- ▶ In equilibrium, any club who is able to attract workers must be such that $V_u(w, \theta, f) \geq V_u$.
- ▶ But since clubs maximize profits, they will not want to offer the unemployed a utility greater than V_u , which is the maximum offered by other clubs.
- ▶ This determines the fee charged by a club which picks w and θ :

$$rV_u(w, \theta, f) = -f + \theta q(\theta)(V_e(w) - V_u(w, \theta, f)),$$

$$rV_u = -f + \theta q(\theta)(V_e(w) - V_u(w, \theta, f)),$$

$$f(w, \theta) = \theta q(\theta)V_e(w) - (r + \theta q(\theta))V_u. \quad (2)$$

2. Search Clubs

Equilibrium (continued)

- ▶ Value of posting a vacancy at a club with (w, θ, g) is $V_v(w, \theta, g)$ such that

$$rV_v(w, \theta, g) = -c - g + q(\theta)(J(w) - V_v(w, \theta, g)),$$

- ▶ where $J(w)$ is the value to the firm of a position with wage w , i.e.

$$rJ(w) = y - w - sJ(w).$$

- ▶ In equilibrium it must be that $V_v(w, \theta, g) \geq 0$
- ▶ Profit maximization by clubs will imply $V_v(w, \theta, g) = 0$,
- ▶ Therefore $V_v(w, \theta, g) = 0$, implying

$$g(w, \theta) = -c + q(\theta)J(w). \quad (3)$$

2. Search Clubs

Equilibrium (continued)

- ▶ Let's look at competition between clubs
- ▶ The profit of a club, per unemployed member, is given by $\pi(w, \theta) = f(w, \theta) + \theta g(w, \theta)$.
- ▶ Free entry of clubs imply that
 - × This cannot exceed zero in equilibrium, for *any* potential club offering w and θ . Therefore :

$$f(w, \theta) + \theta g(w, \theta) \leq 0, \forall w, \theta.$$

- × Profits must be strictly zero for clubs that are nonempty in equilibrium.

2. Search Clubs

Equilibrium (continued)

- Compute $\pi(w, \theta)$ using (2) and (3)

$$f(w, \theta) = \theta q(\theta) V_e(w) - (r + \theta q(\theta)) V_u \quad (2)$$

$$g(w, \theta) = -c + q(\theta) J(w) \quad (3)$$

- to get

$$\pi(w, \theta) = \theta q(\theta) (V_e(w) + J(w) - V_u) - (c\theta + rV_u).$$

× $\theta q(\theta) (V_e(w) + J(w) - V_u)$: benefit of the club, per unemployed member, in terms of job creation.

× $(c\theta + rV_u)$: cost of search per unemployed member.

- $V_e(w) + J(w) - V_u$ = Total net surplus of a match
- $rV_e(w) = w + s(V_u - V_e(w))$ implies $V_e = \frac{w+sV_u}{r+s}$
- $rJ(w) = y - w - sJ(w)$ implies $J = \frac{y-w}{r+s}$
- Therefore, $V_e(w) + J(w) - V_u = \frac{y-rV_u}{r+s}$

2. Search Clubs

Equilibrium (continued)

- ▶ Since $V_e(w) + J(w) - V_u = \frac{y-rV_u}{r+s}$, $\pi(w, \theta) = \pi(\theta)$ does not depend on w .
 - × This is due to the fact that w is a transfer between the firm and the worker, which leaves the total surplus of the match unchanged.
 - × Any increase in w would be matched by an increase in f and a fall in g to maintain V_u and V_v unchanged.

2. Search Clubs

Equilibrium (continued)

- ▶ The preceding equations imply that the equilibrium value of θ, θ^{cs} satisfies

$$\theta^{cs} = \arg \max \pi(\theta) = \theta q(\theta) \frac{y - rV_u}{r + s} - c\theta - rV_u.$$

- ▶ Furthermore,

$$\pi(\theta^{cs}) = 0.$$

- ▶ These two equations jointly determine θ_E and V_u in equilibrium.
- ▶ Note : There is only one type of club in equilibrium.
- ▶ Working through those equations yields the equilibrium θ^{cs} , which must satisfy

$$y(q(\theta^{cs}) + \theta^{cs} q'(\theta^{cs})) = c(r + s - (\theta^{cs})^2 q'(\theta^{cs})). \quad (4)$$

3. Comparing equilibrium and optimal value of θ

θ^{cs} and θ^{so}

- ▶ Competitive Search Equilibrium θ^{cs} is given by

$$y(q(\theta^{cs}) + \theta^{cs} q'(\theta^{cs})) = c(r + s - (\theta^{cs})^2 q'(\theta^{cs})). \quad (4)$$

- ▶ Socially optimal one θ^{so} ?
- ▶ Recall from Lecture 10 that Social Planner FOC are

$$c = \lambda_t \frac{\partial m(u, v)}{\partial v} \quad (5)$$

$$r\lambda_t = y - s\lambda_t - \lambda_t \frac{\partial m}{\partial u} + \dot{\lambda}_t \quad (6)$$

3. Comparing equilibrium and optimal value of θ

Computing θ^{so}

- ▶ We can re-express those conditions in terms of θ .
- ▶ Note that

$$m(u, v) = u\theta q(\theta).$$

- ▶ Therefore,

$$\begin{aligned}\frac{\partial m(u, v)}{\partial v} &= u \times [q(\theta) + \theta q'(\theta)] \frac{\partial \theta}{\partial v} \\ &= q(\theta) + \theta q'(\theta).\end{aligned}$$

- ▶ Hence

$$\lambda_t = \frac{c}{q(\theta_t) + \theta_t q'(\theta_t)}. \quad (7)$$

- ▶ Also,

$$\begin{aligned}\frac{\partial m(u, v)}{\partial u} &= \theta q(\theta) + u[q(\theta) + \theta q'(\theta)] \frac{\partial \theta}{\partial u} \\ &= \theta q(\theta) - \theta[q(\theta) + \theta q'(\theta)] \\ &= -\theta^2 q'(\theta).\end{aligned} \quad (8)$$

3. Comparing equilibrium and optimal value of θ

Computing θ^{so} (continued)

- Substituting (7) and (8) into (6) we get the evolution of θ^{so} :

$$\begin{aligned} \frac{rc}{q(\theta) + \theta q'(\theta)} &= y - \frac{sc}{q(\theta) + \theta q'(\theta)} + \frac{c\theta^2 q'(\theta)}{q(\theta) + \theta q'(\theta)} \\ &+ \frac{d}{dt} \left[\frac{c}{q(\theta_t) + \theta_t q'(\theta_t)} \right]. \end{aligned}$$

- This equation only involves θ and is unstable. Therefore the only non explosive solution is the stationary one, i.e.

$$(r + s - \theta^2 q'(\theta))c = (q(\theta) + \theta q'(\theta)) y. \quad (9)$$

- This determines the optimal $\theta = \theta^{so}$, which is constant over time, like the equilibrium one.

3. Comparing equilibrium and optimal value of θ

$$\theta^{cs} = \theta^{so}$$

- Competitive search :

$$y(q(\theta^{cs}) + \theta^{cs} q'(\theta^{cs})) = c(r + s - (\theta^{cs})^2 q'(\theta^{cs})). \quad (4)$$

- Optimum :

$$y(q(\theta^{so}) + \theta^{so} q'(\theta^{so})) = c(r + s - (\theta^{so})^2 q'(\theta^{so})). \quad (9)$$